CLASS X MATHEMATICS • CHAPTER 11

Areas Related to Circles

"Exhaustive point-by-point presentation covering every concept, formula derivation, and geometric application directly from the NCERT textbook."

Akash Srivastva (www.akashmaths.online)

Sector of a Circle (Minor vs Major)

Formal NCERT Definition

The portion (or part) of the circular region enclosed by **two radii** and their corresponding **arc** is called a **sector of the circle**.

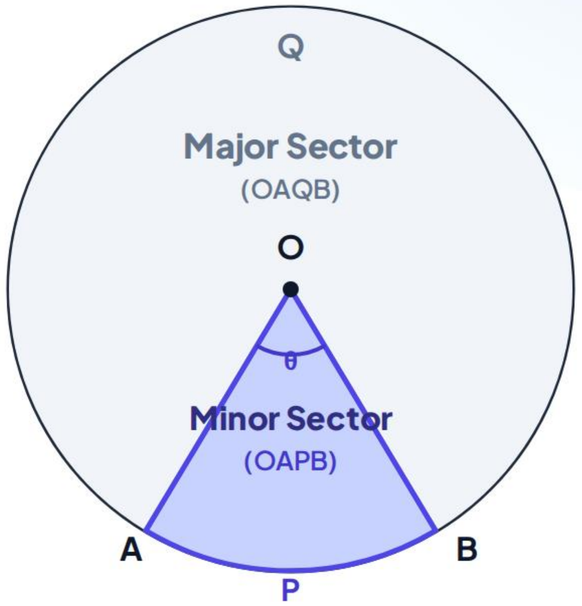
Minor Sector

Enclosed by radii OA, OB and minor arc APB. Central angle = **θ** (where θ < 180°).

Major Sector

Enclosed by radii OA, OB and major arc AQB. Angle = **(360° − θ)**.

Standard Textbook Remark: When we write simply '*sector*', it always implies the **minor sector**, unless specified otherwise.



⚡ Quick Box • Sector Fundamentals

Q1 What two geometric boundaries enclose a sector of a circle? Can a chord form a sector?

CONCEPT RECALL

Q2 If the central angle of a minor sector is 60°, what is the exact degree measure of the angle of the corresponding major sector?

MENTAL MATH

Q3 What is the special name given to a sector whose central angle θ is exactly equal to 90°? What about 180°?

OBSERVATION

Segment of a Circle (Minor vs Major)

Formal NCERT Definition

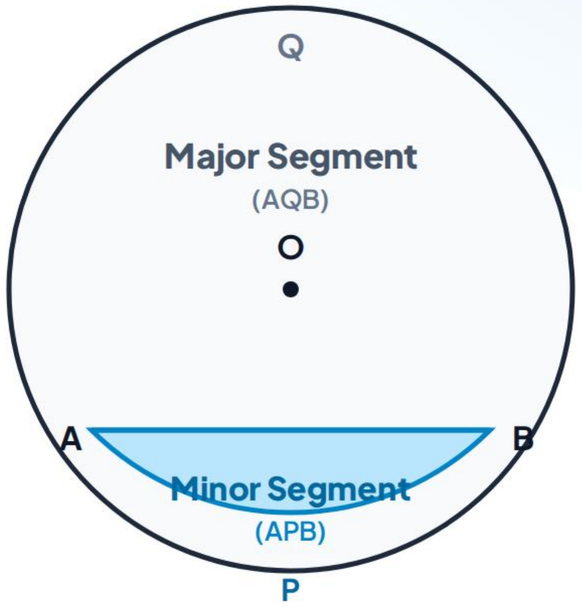
The portion (or part) of the circular region enclosed between a **chord** and its corresponding **arc** is called a **segment of the circle**.

Minor Segment

Region APB enclosed between chord AB and minor arc APB (smaller area).

Major Segment

Region AQB enclosed between chord AB and major arc AQB (larger area).



FUNDAMENTAL AREA CONSERVATION

$$\text{Area(Major Segment)} = \pi r^2 - \text{Area(Minor Segment)}$$

⚡ Quick Box • Segment & Sector Distinction

Q1 What is the fundamental difference in boundary between a sector and a segment? — CORE DISTINCTION

Q2 If a chord passes directly through the centre O (i.e. is a diameter), what happens to the minor and major segments? — SPECIAL CASE

Q3 True or False: "Every segment of a circle is completely contained inside a corresponding sector." Explain why or why not. — SPATIAL REASONING

Page 155 (Textbook)

Derivation: Area of a Sector of a Circle

Step-by-Step Unitary Method

- ✓ A full circular disc can be considered a sector forming a **360°** central angle at centre O, with total area = **πr²**.
- ✓ When central angle is **360°**, Area = **πr²**.
- ✓ When central angle is **1°**, Area = **(πr² / 360)**.
- ✓ Therefore, when central angle is **θ**, Area = **(θ / 360) × πr²**.

KEY FORMULA • AREA OF SECTOR

Area of Sector = (θ / 360) × πr²

Major Sector Area Formula

Two direct approaches to compute the major sector area:

Method 1: Area = πr² – Area(Minor Sector)
Method 2: Area = [(360 – θ) / 360] × πr²

⚡ Quick Box • Sector Area Rapid Drills

Q1 If $\theta = 90°$ (a quadrant of a circle), what fraction of the total circle's area πr^2 is the sector's area? RAPID FRACTION

Q2 For a circle of radius 6 cm and central angle $\theta = 60°$, state the expression for sector area without calculating the final decimal. FORMULA SETUP

Q3 If the radius of a circle is doubled while keeping the central angle θ constant, by what factor does the area of the sector increase? SCALING PROPERTY

Length of an Arc of a Sector

Derivation via Unitary Method

- ✓ The complete boundary perimeter of a circle (subtending **360°** at O) is the circumference = **2πr**.
- ✓ For an angle of **360°**, the arc length is **2πr**.
- ✓ For an angle of **1°**, the arc length is **(2πr / 360)**.
- ✓ Hence, for a sector angle **θ**, the arc length **l** is:

ARC LENGTH FORMULA

Length of Arc (l) = (θ / 360) × 2πr

Total Perimeter of a Sector

Students frequently confuse *arc length* with the *total perimeter of a sector*. The entire perimeter includes both bounding radii!

Perimeter of Sector = l + 2r = [(θ / 360) × 2πr] + 2r

Useful Sector Area Relation

Notice that Area = $(\theta / 360) \times \pi r^2 = \frac{1}{2} \times [(\theta / 360) \times 2\pi r] \times r$:

Area of Sector = ½ × l × r

⚡ Quick Box • Arc Length & Sector Perimeter

Q1 What is the length of the arc of a semicircle of radius r? What is the total perimeter of that semicircle?

COMMON ERROR CHECK

Q2 If a sector has arc length l = 10 cm and radius r = 7 cm, what is the area of this sector using the relation Area = ½ × l × r?

QUICK CALCULATION

Q3 Which is larger: the arc length of a 60° sector in a circle of radius 14 cm, or a 120° sector in a circle of radius 7 cm?

RATIO COMPARISON

Area of a Segment of a Circle (Strategy)

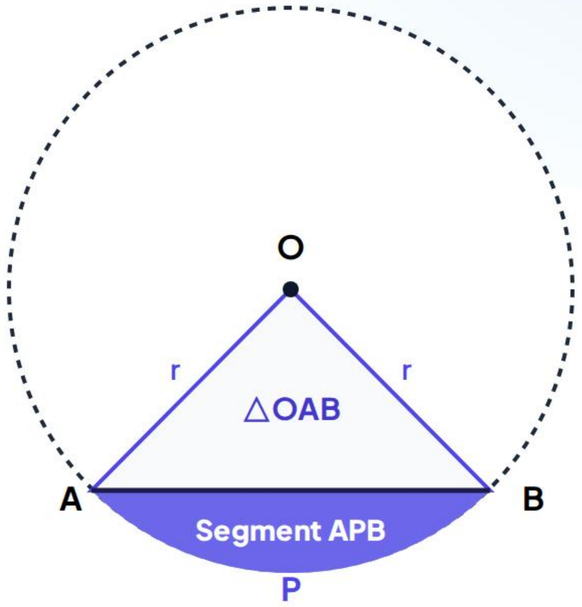
Master Subtraction Principle

The area of minor segment **APB** is obtained by subtracting the area of the corresponding triangle △**OAB** from the area of sector **OAPB**:

MASTER SEGMENT EQUATION

Area(Segment APB) = Area(Sector OAPB) − Area(△OAB)

= [(θ / 360) × πr²] − Area(△OAB)



Area of Major Segment AQB

Area(Major Segment) = πr² − Area(Minor Segment APB)

⚡ Quick Box • Segment Strategy Check

Q1 Why can't we find the area of a segment directly with a simple one-step unitary formula like the sector area?

GEOMETRIC INSIGHT

Q2 If the area of minor segment APB is calculated as 35 cm^2 in a circle of total area 154 cm^2, what is the exact area of the major segment?

MENTAL MATH

Q3 In the formula Area(Segment) = Area(Sector) − Area(△OAB), what always happens to the two sides OA and OB of △OAB?

TRIANGLE PROPERTY

Pages 156–158 (Textbook)

Finding Area of △OAB for Key Central Angles

Case 1: θ = 60°

Since OA = OB = r, ∠OAB = ∠OBA = (180° − 60°)/2 = 60°.

Hence, △OAB is Equilateral!

EQUILATERAL AREA

$$(\sqrt{3} / 4) \times r^2$$

Case 2: θ = 90°

OA ⊥ OB at centre O. Radii act as perpendicular base and altitude.

Right-angled at vertex O!

RIGHT TRIANGLE AREA

$$\frac{1}{2} \times r \times r = \frac{1}{2} r^2$$

Case 3: θ = 120°

Draw OM ⊥ AB. In △OMA (30°-60°-90°): OM = r cos 60° = r/2, AM = r sin 60° = r√3/2, AB = r√3.

TRIG / ALTITUDE RESULT

$$\frac{1}{2} \times AB \times OM = (\sqrt{3} / 4) r^2$$

Universal Trigonometric Shortcut: For any central angle θ, Area of △OAB = $\frac{1}{2} r^2 \sin\theta$. (Notice sin 90° = 1 → ½ r²; sin 60° = sin 120° = √3/2 → √3/4 r²).

⚡ Quick Box • Triangle Area Rapid Check

Q1 If a chord subtends 90° at the centre of a circle of radius 10 cm, what is the area of triangle OAB in just 5 seconds?

MENTAL CALCULATION

Q2 Why does a 120° central angle yield the exact same triangle area as a 60° central angle for circles of the same radius?

SYMMETRY / TRIG INSIGHT

Q3 When $\theta = 60°$, why are all three sides of $\triangle OAB$ guaranteed to be equal to radius r?

GEOMETRY JUSTIFICATION

Real-World Circular Models in Exercise 11.1

Clock Minute Hand

60 min = 360° rotation.
1 min = 6° rotation.
In 5 min, angle swept = **30°**.

Q3 Model

Tethered Horse

Corner of square field = **90°** angle.
Grazing area forms a **quadrant of radius r** (rope length).

Q8 Model

Symmetrical Ribs

Umbrella with 8 flat ribs divides circle into 8 equal sectors: **θ = 360°/8 = 45°**.
Brooch (10 sectors) = **36°**.

Q9 & Q10 Models

Beams & Wipers

Lighthouse beam: θ = 80°, r = 16.5 km.
Car wipers: 2 non-overlapping blades sweeping θ = 115°.

Q11 & Q12 Models

Teacher's Tip: In all real-world contextual questions, identify two key parameters first: **Radius (r)** and **Sector Angle (θ)**.

⚡ Quick Box • Real-World Context Prompts

Q1 Through how many degrees does the minute hand of a clock turn between 12:00 PM and 12:20 PM?

CLOCK GEOMETRY

Q2 If a round brooch has 5 diameters dividing it into equal sectors, how many total sectors are formed and what is each central angle?

SYMMETRY ANALYSIS

Q3 A horse's tether rope is increased from 5 m to 10 m in the corner of a field. Does the grazing area double? Explain why or why not.

CRITICAL THINKING

Worked Highlights: Clock Hand & Grazing Horse

EXERCISE 11.1 • QUESTION 3

Area Swept by Minute Hand in 5 Minutes

Given: Length of minute hand r = 14 cm, time = 5 min.

- ✓ Angle described in 60 min = 360° ⇒ Angle in 5 min: θ = (360° / 60) × 5 = **30°**.
- ✓ Area swept = (θ / 360) × πr² = (30 / 360) × (22 / 7) × 14 × 14
- ✓ = (1 / 12) × 22 × 2 × 14 = (1 / 12) × 616 = **154 / 3 cm²**

Final Answer: 154/3 cm² (≈ 51.33 cm²)

EXERCISE 11.1 • QUESTION 8

Horse Tied to Corner of Square Field

Given: Square field corner (θ = 90°), rope r_1 = 5 m, extended r_2 = 10 m.

- ✓ (i) Initial area = (90/360) × 3.14 × 5² = ¼ × 3.14 × 25 = **19.625 m²**.
- ✓ (ii) Extended area = ¼ × 3.14 × 10² = ¼ × 314 = **78.5 m²**.
- ✓ Increase in area = 78.5 – 19.625 = **58.875 m²**.

Increase: 58.875 m²

Worked Highlight: Segment with 120° Central Angle

EXAMPLE 2 / EXERCISE 11.1 Q7 TYPE

Area of Segment AYB with r = 21 cm, θ = 120°

Given: Radius r = 21 cm, central angle θ = 120°, π = 22/7.

Step 1: Area of Sector OAYB

Area(Sector) = (120/360) × (22/7) × 21 × 21 = ⅓ × 22 × 3 × 21 = **462 cm²**

Step 2: Area of Triangle OAB (OM ⊥ AB)

OM = 21 cos 60° = 21/2 cm | AB = 2 × 21 sin 60° = 21√3 cm
Area(△OAB) = ½ × (21√3) × (21/2) = **(441√3 / 4) cm²**

Step 3: Segment Subtraction & Exact Result

Area(Segment AYB) = Area(Sector OAYB) – Area(△OAB)

EXACT TEXTBOOK EXPRESSION

[462 – (441√3 / 4)] cm² = (21 / 4)(88 – 21√3) cm²

If evaluated with √3 = 1.732:

441√3 / 4 ≈ 441 × 1.732 / 4 ≈ 190.95 cm²

Area of segment ≈ 462 – 190.95 = **271.05 cm²**

Page 160 (Textbook)

Chapter 11 Master Formula Checklist

Core Formulas (Mandatory Recall)

- ✓ **Arc Length of Sector (l):**
  $l = (\theta / 360) \times 2\pi r$
- ✓ **Area of Sector of Angle θ:**
  $\text{Area} = (\theta / 360) \times \pi r^2 = \frac{1}{2} l r$
- ✓ **Area of Segment of a Circle:**
  $\text{Area(Segment)} = \text{Area(Sector)} - \text{Area}(\triangle OAB)$

Strategic Rules of Thumb

- ✓ **Major Sector Area:** πr^2 – Area(Minor Sector) or $[(360 - \theta)/360] \times \pi r^2$.
- ✓ **Major Segment Area:** πr^2 – Area(Minor Segment).
- ✓ **Triangle Area Shortcuts:**
  $\theta = 60°$: $(\sqrt{3}/4)\, r^2$ | $\theta = 90°$: $\frac{1}{2} r^2$ | $\theta = 120°$: $(\sqrt{3}/4)\, r^2$.
- ✓ **Standard π Convention:** Use $\pi = 22/7$ unless explicitly instructed to take $\pi = 3.14$.

NCERT CLASS X MATHEMATICS

Thank You!

"Geometry is knowledge of the eternally existent. Master the circle, and you master the harmony of the universe."

Akash Srivastva (www.akashmaths.online)

Designed for high-impact classroom presentation and print-quality educational delivery.